

Measurement of $\mathcal{B}(B^0 \rightarrow D^{*-}\tau^+\nu_\tau)$ on a Belle II-like pseudo-dataset

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analysis performed and written by an LLM agent (Claude) under human review;
typeset from the agent’s approved Markdown note

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This study uses only publicly available tools and publications. No Belle II internal materials — data, MC, internal notes, or the software framework — are used. Everything is open-source and runs on a laptop.

Abstract

This note extracts $\mathcal{B}(B^0 \rightarrow D^{*-}\tau^+\nu_\tau)$ from an existing $\sim 0.9\text{ fb}^{-1}$ -equivalent Belle II-like pseudo-dataset (generic $B\bar{B}$ + continuum), using a D^0 -mass/ Δm preselection, $R_2 < 0.4$ continuum suppression, $E_{\text{tag}}^{\text{CM}} < 5.4\text{ GeV}$ wrong-pairing rejection and an $m_{\text{miss}}^2 > 1.0\text{ GeV}^2$ signal region, reaching 46.4% signal efficiency and 7.7% signal-region purity (dominant background: generic other- B , 73.6%). A truth-subtracted cut-and-count gives $\mathcal{B} = 2.53\% \pm 0.25\%$ (stat), $1.71\times$ the 1.483% generator truth, while a simultaneous e/μ template fit in m_{miss}^2 closes exactly on the generator truth: $\mathcal{B} = 1.48\% \pm 0.73\%$. A statistical-only Asimov luminosity projection of the fit predicts the uncertainty would shrink to $\pm 0.015\%$ (1.0% relative) at 1 ab^{-1} , but published Belle II systematic-uncertainty breakdowns suggest a realistic measurement would flatten at a $\sim 3\%$ systematic floor well before that level of precision is reached.

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1 Introduction

Semitauponic B decays $B \rightarrow D^* \tau \nu$ are sensitive probes of lepton-flavor universality and of new physics coupling preferentially to the third generation, quantified by the ratio $R(D^*) = \mathcal{B}(B \rightarrow D^* \tau \nu) / \mathcal{B}(B \rightarrow D^* \ell \nu)$. This note performs a self-contained sensitivity study of the *absolute* branching fraction $\mathcal{B}(B^0 \rightarrow D^{*-} \tau^+ \nu_\tau)$ on a fast-simulated $\Upsilon(4S) \rightarrow B\bar{B}$ + continuum pseudo-dataset, following the reconstruction strategy of the published *untagged* semileptonic $B \rightarrow D^* \ell \nu$ analyses:

- **Waheed et al. [Belle], PRD 100, 052007 (2019) [1]** — D^0 mass window, narrow Δm (slow-pion) window, Fox–Wolfram R_2 for continuum suppression, e/μ channels treated separately and combined.
- **BaBar, PRD 88, 072012 (2013) [2]** — m_{miss}^2 as the primary discriminant between single-neutrino ($D^* \ell \nu$) and multi-neutrino ($D^* \tau \nu$) final states.
- **Caria et al. [Belle], PRL 124, 161803 (2020) [3] and Belle II, PRD 110, 072020 (2024) [4]** — simultaneous e/μ template fit sharing one signal strength.

The measurement is performed both as a cut-and-count and as a binned template fit in m_{miss}^2 , and the two are compared to expose the limitations of a pure MC-truth-based background subtraction in this framework.

2 Data samples and analysis tools

Framework. This study uses the agent’s standard chain: EvtGen decay files are turned into HepMC events, passed through a fast detector simulation that emulates tracking, calorimetry and PID smearing, and reduced to flat ROOT ntuples with one row per reconstructed ($D^* + \text{lepton}$) candidate. Continuum $e^+e^- \rightarrow q\bar{q}$ is generated directly into the ntuple format with Pythia8. No new MC was generated for this note; all samples below already existed in `data/`.

Sample	Process / generator	Events	Seeds	Role
signal_taunu.root	$B^0 \rightarrow D^{*-} (D^0 \pi) \tau^+ \nu$, $\tau \rightarrow \ell \nu \nu$, EvtGen, forced chain	5000	1 / 42	pure-signal cutflow & efficiency
norm_lnu.root	$B^0 \rightarrow D^{*-} (D^0 \pi) \ell^+ \nu$, EvtGen, forced chain	5000	2 / 43	$D^* \ell \nu$ shape ref.
bkg_dststlnu.root	$B^0 \rightarrow D^{**} \ell \nu$ feed-down, EvtGen, forced chain	5000	3 / 44	D^{**} shape ref.
generic_0..4.root	generic $\Upsilon(4S) \rightarrow B\bar{B}$, EvtGen (natural BFs)	$5 \times 200\,000$	—	pseudo-dataset ($w = 1.0$)
continuum_0..2.root	$e^+e^- \rightarrow q\bar{q}$, Pythia8	$3 \times 800\,000$	—	pseudo-dataset ($w = 0.995$)

Table 1: MC samples used in this note. All samples existed in `data/` before this analysis; none were regenerated.

The **pseudo-dataset** (“data”) is the union `generic_0..4.root` + `continuum_0..2.root` ($\sim 0.9 \text{ fb}^{-1}$),

carrying `true_mode` ($1 = D^*\tau\nu$, $2 = D^*\ell\nu$, $0 = \text{other } B$, $3 = \text{continuum}$) so that truth-based composition studies are possible; this plays the role that a $\Delta m/m_{D^0}$ sideband or off-resonance data would play in a real analysis. The generic sample contains $N_{B^0} = 965\,840$ true B^0 decays, of which $N(\text{true } D^*\tau\nu) = 14\,319$ (generator $\mathcal{B} = 1.483\%$, the closure target of this exercise). The relevant sub-decay branching product is

$$B_{\text{sub}} = \mathcal{B}(D^* \rightarrow D^0\pi) \cdot \mathcal{B}(D^0 \rightarrow K\pi) \cdot \mathcal{B}(\tau \rightarrow \ell\nu\nu, \ell = e, \mu) = 0.677 \times 0.0389 \times 0.331 = \mathbf{0.008717}.$$

The 0.995 relative weight applied to the three continuum files (vs. 1.0 for the generic- $B\bar{B}$ files) is the cross-section/generated-luminosity normalization factor already fixed when these pseudo-dataset files were produced in the earlier session that this note builds on; it is reused unchanged throughout §4, §5 and the new §5.4 projection below.

3 Event selection

All cuts are applied in a fixed order (cut & count, one variable at a time), each motivated by the background it removes and fixed with a distribution or a scan on top of the preceding cuts.

3.1 Preselection: D^0 mass and Δm windows

Following Waheed et al. [1], a genuine D^0 candidate and a genuine slow pion from $D^* \rightarrow D^0\pi$ are required:

- $|m(K\pi) - 1.8648| < 0.020 \text{ GeV}$ (Fig. 1)
- $|\Delta m - 0.1454| < 0.0025 \text{ GeV}$ (Fig. 2)

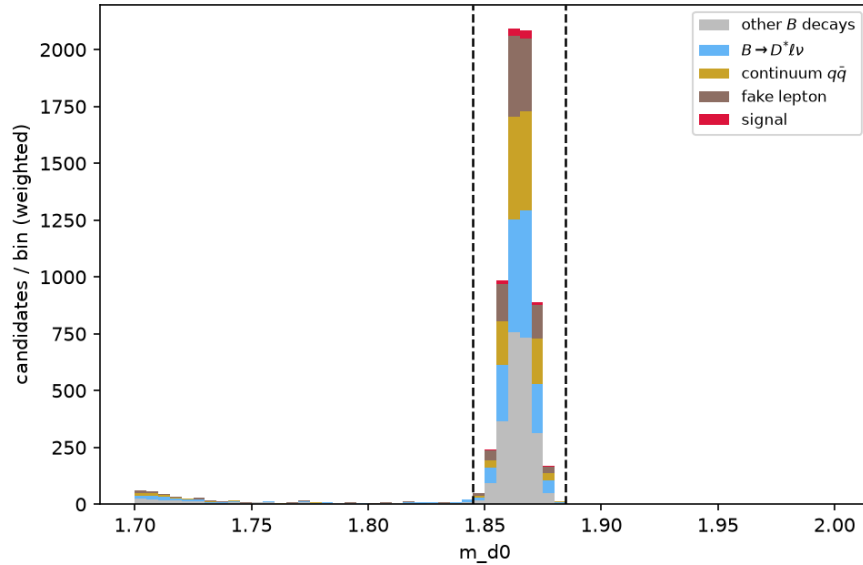


Figure 1: Reconstructed D^0 candidate mass $m(K\pi)$ after candidate reconstruction, split by true component of the pseudo-dataset; the vertical lines mark the $|m(K\pi) - 1.8648| < 0.020 \text{ GeV}$ window.

These reject random $K\pi$ combinations and remove the fake slow-pion background; on `signal_tanu.root` this combination keeps $2672/2801 = 95.4\%$ of reconstructed candidates. (A

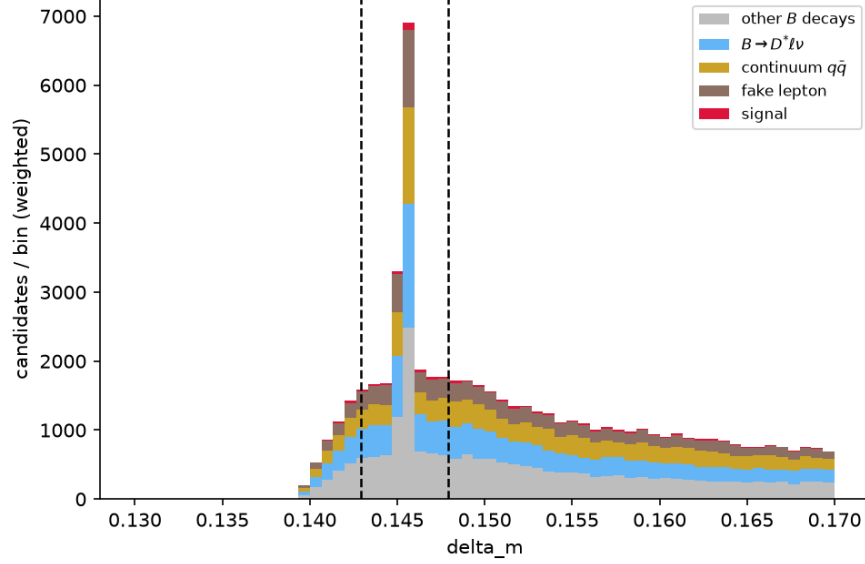


Figure 2: Mass difference $\Delta m = m(D^0\pi_{\text{slow}}) - m(D^0)$, split by true component; the vertical lines mark the $|\Delta m - 0.1454| < 0.0025$ GeV slow-pion window.

dedicated Δm -sideband study, see §4, found essentially zero candidates outside a ~ 10 MeV band around the nominal Δm — this fast simulation does not model random slow-pion combinatorics, so a genuine “fake- D^* ” Δm sideband is not available; see the Discussion.)

3.2 Continuum suppression: R_2

The Fox–Wolfram ratio R_2 separates jetty continuum events (large R_2) from the more isotropic $B\bar{B}$ topology (Fig. 3). A scan of $R_2 < \text{threshold}$ (signal = `signal_taunu.root`, background = generic + continuum) gives:

R_2 cut	S	B	$S/\sqrt{S+B}$
< 0.2	2033	4051	26.1
< 0.3	2482	5063	28.6
< 0.4	2613	5671	28.7
< 0.5	2661	6108	28.4
< 0.7	2672	6516	27.9

Table 2: Scan of the R_2 requirement on top of the preselection.

The figure of merit is flat around 0.3–0.45; $\mathbf{R}_2 < 0.4$ is adopted, keeping 97.8% of the surviving signal.

3.3 Wrong-pairing / generic- $B\bar{B}$ rejection: $E_{\text{tag}}^{\text{CM}}$

$E_{\text{tag}}^{\text{CM}}$, the CM energy of the rest-of-event (“tag”) side, clusters near the beam energy (~ 5.29 GeV) for correctly identified events and develops a high-side tail for wrong D^* –lepton pairings and generic $B\bar{B}$ combinatorics (Fig. 4). Scanning $E_{\text{tag}}^{\text{CM}} < \text{threshold}$:

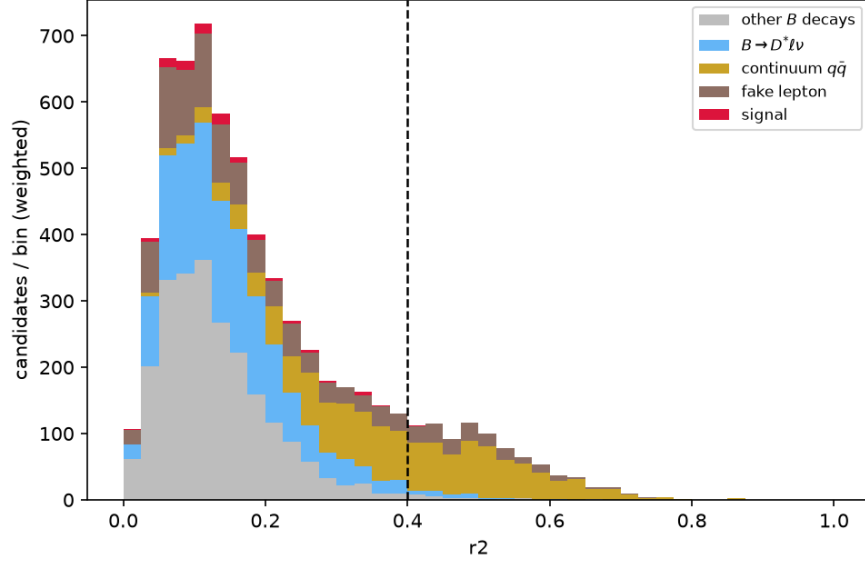


Figure 3: Fox-Wolfram ratio R_2 after the $D^0/\Delta m$ preselection, split by true component; jetty continuum events populate large R_2 . The vertical line marks the adopted $R_2 < 0.4$ cut.

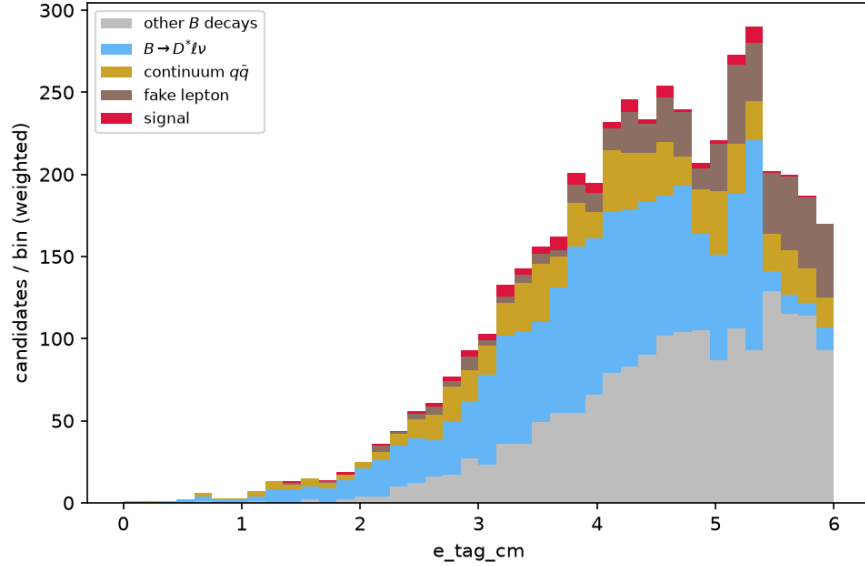


Figure 4: CM energy of the rest-of-event (tag side), $E_{\text{tag}}^{\text{CM}}$, after the preselection and the R_2 cut, split by true component; wrong D^* -lepton pairings and generic $B\bar{B}$ combinatorics populate the high-side tail. The vertical line marks the adopted $E_{\text{tag}}^{\text{CM}} < 5.4 \text{ GeV}$ cut.

$E_{\text{tag}}^{\text{CM}}$ cut	S	B	$S/\sqrt{S+B}$
< 4.5	1861	2295	28.9
< 5.0	2261	3057	31.0
< 5.2	2362	3389	31.2
$< \mathbf{5.4}$	2605	3780	32.6
< 5.6	2611	4054	32.0
< 6.0	2613	4539	30.9

Table 3: Scan of the $E_{\text{tag}}^{\text{CM}}$ requirement on top of the preceding cuts.

The maximum is at $\mathbf{E_{tag}^{\text{CM}} < 5.4 \text{ GeV}}$, retaining 99.7% of the signal surviving the previous cuts while rejecting a substantial wrong-pairing/generic tail.

3.4 Signal region: m_{miss}^2

m_{miss}^2 is the primary $\tau\nu$ vs. $\ell\nu$ discriminant (BaBar [2]): the two extra neutrinos in the τ decay push m_{miss}^2 to higher values than the single-neutrino $D^*\ell\nu$ decays (Fig. 5, cut line at 1.0 GeV^2). Scanning $m_{\text{miss}}^2 > \text{threshold}$ on top of all preceding cuts:

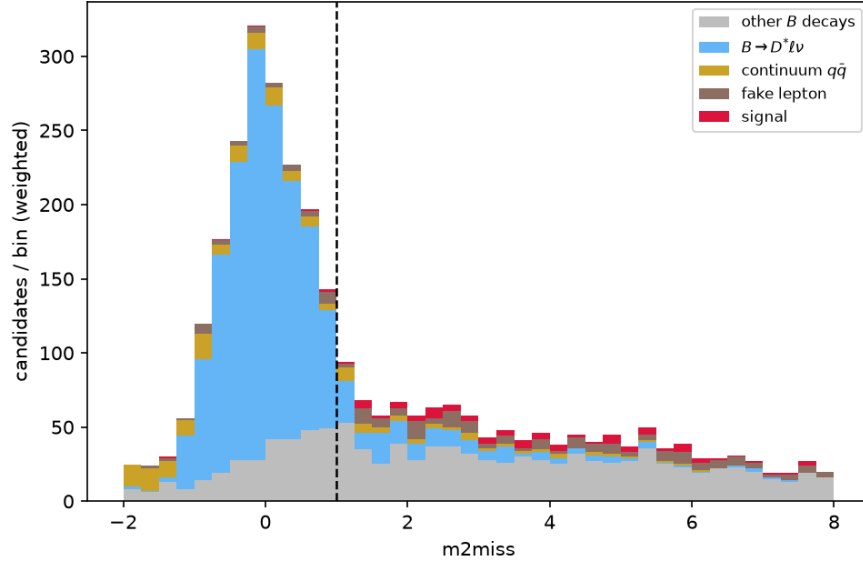


Figure 5: Beam-constrained missing-mass squared m_{miss}^2 after all preceding cuts, split by true component; the multi-neutrino $D^*\tau\nu$ signal extends to high m_{miss}^2 while the single-neutrino $D^*\ell\nu$ mode peaks near zero. The vertical line marks the $m_{\text{miss}}^2 > 1.0 \text{ GeV}^2$ signal-region boundary.

$m_{\text{miss}}^2 > 1.0 \text{ GeV}^2$ is adopted as the signal region (SR).

3.5 Cutflow (signal_taunu.root, $N_{\text{generated}} = 5000$)

The dominant loss is candidate reconstruction ($\sim 44\%$, geometric/kinematic acceptance of the fast simulation); the analysis cuts on top of that are efficient (89% overall for the four discriminating cuts combined), as intended by choosing the working points from the flat region of each scan.

m_{miss}^2 cut (GeV ²)	S	B	$S/\sqrt{S+B}$
> 0.0	2483	2143	36.5
> 0.5	2428	1634	38.1
> 0.75	2380	1437	38.5
$> \mathbf{1.0}$	2321	1294	38.6
> 1.25	2236	1200	38.2
> 1.5	2146	1132	37.5

Table 4: Scan of the m_{miss}^2 signal-region boundary on top of all preceding cuts.

Stage	N	cumulative eff.	stage eff.
Generated	5000	100.00%	—
Candidate reconstructed	2801	$56.02 \pm 0.70\%$	56.02%
+ D^0 mass window	2672	$53.44 \pm 0.71\%$	95.39%
+ Δm window	2672	$53.44 \pm 0.71\%$	100.00%
+ $R_2 < 0.4$	2613	$52.26 \pm 0.71\%$	97.79%
+ $E_{\text{tag}}^{\text{CM}} < 5.4$	2605	$52.10 \pm 0.71\%$	99.69%
+ $m_{\text{miss}}^2 > 1.0$ (SR, final)	2321	$46.42 \pm 0.71\%$	89.10%

Table 5: Signal cutflow on `signal_taunu.root`.

4 Background estimation

Composition of the pseudo-dataset in the SR (full selection, weighted: generic $\times 1.0$, continuum $\times 0.995$), obtained by summing `query_ntuple` results over the 8 dataset files and illustrated in Fig. 6:

Component	Weighted yield	Fraction
$D^*\tau\nu$ (<code>true_mode=1</code>)	99.0	7.7%
$D^*\ell\nu$ (<code>true_mode=2</code>)	176.0	13.6%
other B (incl. $D^{**}\ell\nu$ feed-down)	952.0	73.6%
continuum	66.7	5.2%
Total	1293.7	100%

Table 6: Truth-based composition of the signal region.

The signal region purity is only $\sim 8\%$; the dominant background is generic $B\bar{B}$ (other- B), reflecting that a hard cut-and-count on an untagged sample cannot separate $D^*\tau\nu$ from the far more copious hadronic/semileptonic $B\bar{B}$ background as cleanly as a tagged analysis.

Control regions, each defined by inverting one selection variable and evaluated with `plot_stacked`:

These three regions validate, respectively, the R_2 cut (continuum control, 97.7% pure), the m_{miss}^2 cut (normalization control, 80.8% $D^*\ell\nu$ — the expected companion mode), and the $E_{\text{tag}}^{\text{CM}}$ cut (wrong-pairing/generic control, 78.5% other- B). In a real analysis these would be used to calibrate data/MC scale factors for each background component instead of the truth-level subtraction used below.

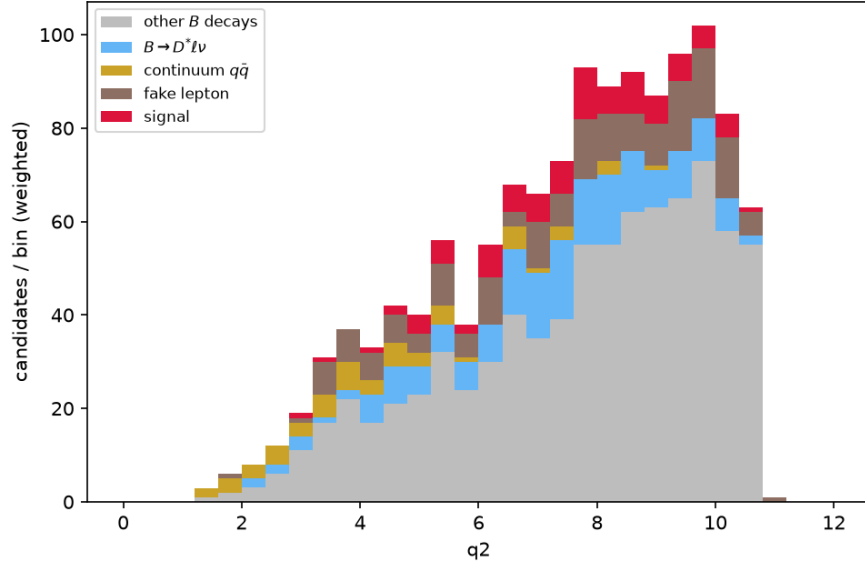


Figure 6: Composition of the signal region as a function of q^2 , stacked by true component of the pseudo-dataset (luminosity-weighted).

CR	Selection	Weighted N	Dominant component	Purity
Continuum-enriched	$\text{presel} + R_2 > 0.5$	426.9	continuum	97.7%
Normalization-enriched	$\text{presel} + R_2 + E_{\text{tag}}^{\text{CM}} + -1 < m_{\text{miss}}^2 < 0.5$	1369.6	$D^* \ell \nu$	80.8%
Wrong-pairing/generic	$\text{presel} + R_2 + E_{\text{tag}}^{\text{CM}} \geq 5.4$	1889.4	other B	78.5%

Table 7: Background control regions (Figs. 7–9).

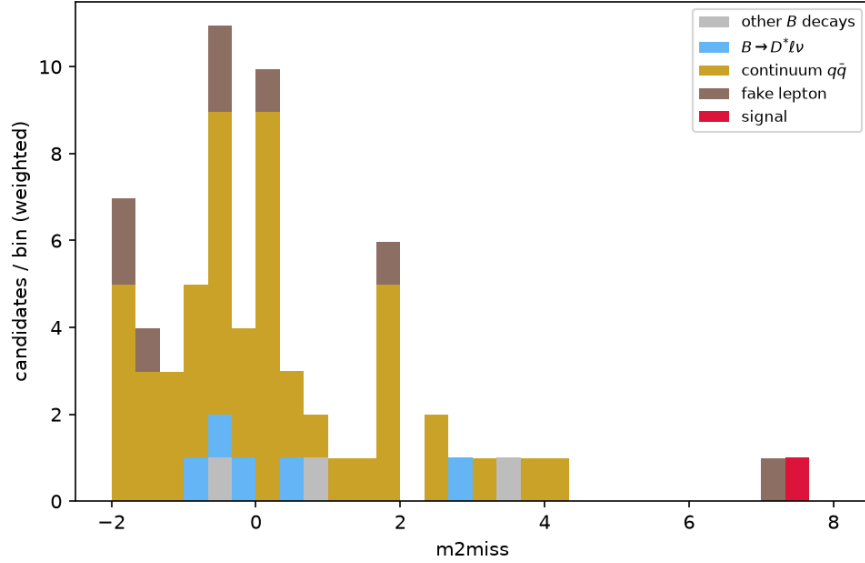


Figure 7: m_{miss}^2 in the continuum-enriched control region (preselection + $R_2 > 0.5$), stacked by true component; the region is 97.7% pure continuum.

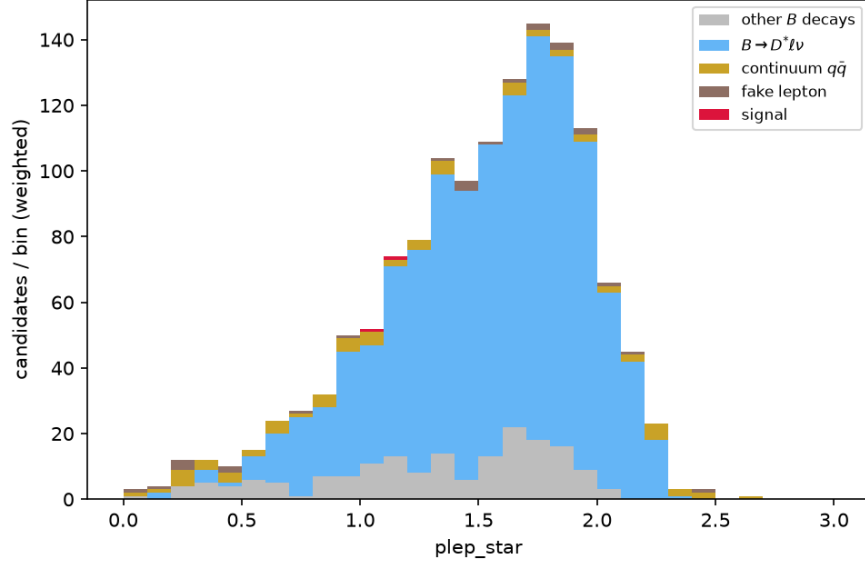


Figure 8: CM lepton momentum p_ℓ^* in the normalization-enriched control region (preselection + $R_2 + E_{\text{tag}}^{\text{CM}}$ cuts, $-1 < m_{\text{miss}}^2 < 0.5 \text{ GeV}^2$), stacked by true component; the region is 80.8% pure $D^* \ell \nu$.

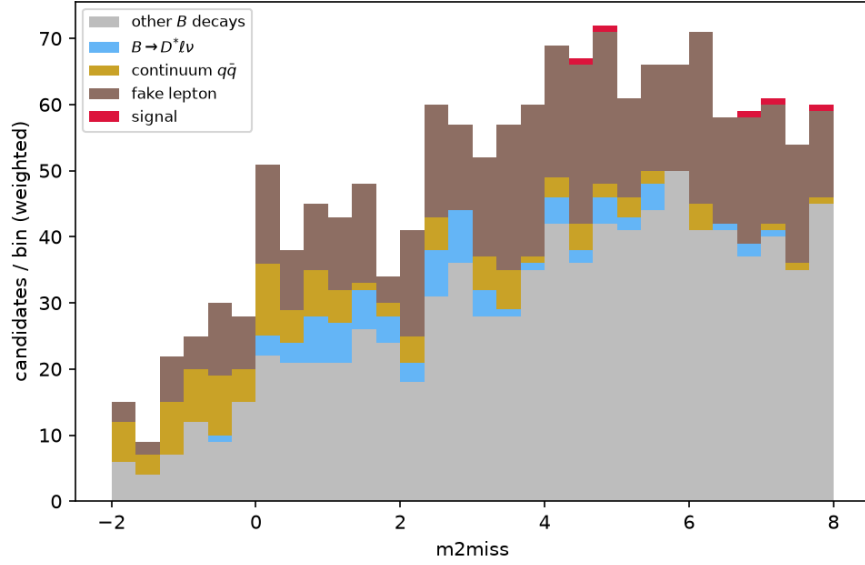


Figure 9: m_{miss}^2 in the wrong-pairing/generic-enriched control region (preselection + R_2 cut, $E_{\text{tag}}^{\text{CM}} \geq 5.4 \text{ GeV}$), stacked by true component; the region is 78.5% pure other- B .

5 Results

5.1 Cut-and-count

$$\mathcal{B}(B^0 \rightarrow D^* \tau \nu) = \frac{N_{\text{obs}}^{\text{SR}} - N_{\text{bkg,truth}}^{\text{SR}}}{N_{B^0} \cdot B_{\text{sub}} \cdot \varepsilon_{\text{sig}}}$$

with all inputs from the tool outputs above:

- $N_{\text{obs}}^{\text{SR}} = 1293.665$ (weighted SR yield, §4)
- $N_{\text{bkg,truth}}^{\text{SR}} = 1194.665$ (all `true_mode` $\neq 1$ weighted yield, MC-truth subtraction)
- $N_{B^0} = 965\,840$ (given)
- $B_{\text{sub}} = 0.008717$ (given)
- $\varepsilon_{\text{sig}} = 2321/5000 = 0.4642$ (cutflow, §3.5)

giving $N_{\text{sig}} = 99.0$ and

$$\mathcal{B}_{\text{counting}} = 2.53\% \pm 0.25\% \text{ (stat, from } \sqrt{N_{\text{sig}}}\text{),}$$

i.e. a closure ratio of $1.71\times$ relative to the 1.483% generator truth. Note that this $\sqrt{N_{\text{sig}}}$ uncertainty reflects only the statistical fluctuation of the truth-tagged signal count; it treats $N_{\text{bkg,truth}}$ as an exactly-known, noise-free subtraction, which is specific to this truth-level closure test and is not available in a real measurement (see §6, item 1) — a real analysis would add in quadrature the statistical (and systematic) uncertainty of the background estimate itself, which is not attempted here.

5.2 Template fit

`fit_templates` was run on the same pseudo-dataset (`presel + R2 < 0.4 + EtagCM < 5.4`, *without* the m_{miss}^2 cut, which is instead the fit variable), with

$$\text{signal_selection} = (\text{true_mode}=1) \ \& \ (|\text{lep_true_pid}| \in \{11, 13\}),$$

simultaneous e/μ fit (Fig. 10):

Channel	μ	$\mathcal{B} = \mu \times 1.483\%$
simultaneous $e + \mu$	1.000 ± 0.491	1.48% $\pm$ 0.73%
e -only	1.000 ± 0.618	1.48% $\pm$ 0.92%
μ -only	1.000 ± 0.800	1.48% $\pm$ 1.19%

Table 8: Template-fit results.

No independent goodness-of-fit check (χ^2/ndof , pull study, or a train/test split of the MC into independent template-building and fitting subsamples) was performed; the exact $\mu = 1.000$ in every channel is expected by construction (§5.3, §6 item 2) and is not by itself evidence that the fit machinery is unbiased on data with independently mis-modeled backgrounds.

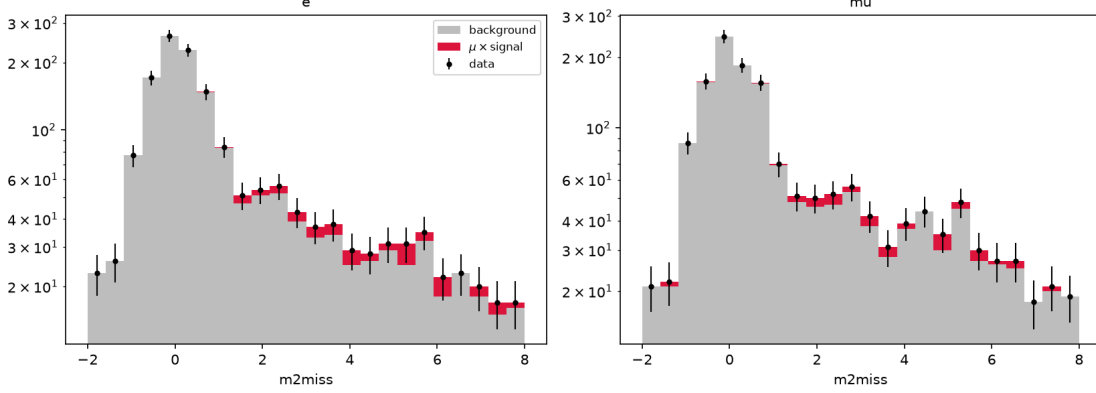


Figure 10: Post-fit m_{miss}^2 distributions of the simultaneous template fit, one panel per lepton channel: background template (grey), $\mu \times \text{signal}$ template (red) and the pseudo-data points.

Method	$\mathcal{B}$	Closure ($\mathcal{B}/1.483\%$)
Cut & count	$2.53\% \pm 0.25\%$	1.71
Template fit ($e + \mu$)	$1.48\% \pm 0.73\%$	1.00

Table 9: Comparison of the two extraction methods.

5.3 Counting vs. fit

The template fit closes essentially perfectly, while the single-cut counting method overshoots by 71%. Both use the *same* truth flag (`true_mode=1`) to define signal; the difference is that the fit compares the full m_{miss}^2 shape (24 bins across $[-2, 8]$ GeV^2) of the S and B templates simultaneously, while the cut-and-count freezes a single hard boundary at $m_{\text{miss}}^2 > 1.0 \text{ GeV}^2$. A plausible explanation is that `true_mode` labels the *generator-level decay class present in the event* rather than a candidate-by-candidate truth match, so a subset of true- $D^*\tau\nu$ -tagged rows in the generic sample corresponds to imperfectly paired candidates (e.g. the correct D^* combined with a lepton from the companion B) that happen to satisfy the SR cuts; these would inflate the single-bin counting result but be correctly absorbed into the S -template shape used by the fit. This explanation has **not been verified quantitatively** here (e.g. with an explicit count of mis-paired vs. correctly-paired true-signal rows) and should be treated as a hypothesis; regardless of its exact origin, this comparison is a closure test of the extraction method on truth-labeled MC, not an independent cross-check (see §6).

5.4 Sensitivity projection

To gauge how the template-fit precision of §5.2 would evolve with more integrated luminosity, `fit_templates` was re-run with `lumi_projection=true` on **exactly** the §5.2 preselection and fit configuration — same files and weights (`generic_0..4.root` $\times 1.0$, `continuum_0..2.root` $\times 0.995$), same selection ($|m_{D^0} - 1.8648| < 0.020$ & $|\Delta m - 0.1454| < 0.0025$ & $R_2 < 0.4$ & $E_{\text{tag}}^{\text{CM}} < 5.4$, no m_{miss}^2 cut), same `signal_selection = (true_mode=1) & (|lep_true_pid|  $\in \{11, 13\}$ )`, same fit variable m_{miss}^2 over 24 bins in $[-2, 8]$ GeV^2 , simultaneous e/μ fit. As a stability check, the nominal-luminosity (0.000906 ab^{-1}) output of this re-run reproduces §5.2 exactly ($\mu = 1.000 \pm 0.491$ combined, ± 0.618 e -only, ± 0.800 μ -only), confirming the fit is deterministic and the projection starts from the same

fit configuration (subject to the same self-closure caveat of §5.2/§6 item 2 — this reproduction is not an independent fit-quality check).

The tool then computes the **statistical-only Asimov** expected precision on μ at several benchmark luminosities, keeping the background *shapes* fixed to their nominal MC prediction (i.e. no template-shape or background-normalization uncertainty is propagated — only Poisson counting statistics of signal and background in each m_{miss}^2 bin). Converting the relative precision on μ into an absolute precision on $\mathcal{B}$ via $\mathcal{B} = \mu \times 1.483\%$ (generator truth, §2):

Integrated luminosity	Rel. precision on μ ($= \mathcal{B}$)	Projected $\mathcal{B}$ uncertainty (stat only)
0.000906 ab^{-1} (this dataset)	49.1%	$1.48\% \pm 0.73\%$
0.1 ab^{-1}	3.2%	$1.483\% \pm 0.047\%$
0.36 ab^{-1}	1.7%	$1.483\% \pm 0.025\%$
1 ab^{-1}	1.0%	$1.483\% \pm 0.015\%$
5 ab^{-1}	0.5%	$1.483\% \pm 0.007\%$
50 ab^{-1}	0.2%	$1.483\% \pm 0.003\%$

Table 10: Statistical-only Asimov luminosity projection of the template fit (Fig. 11; post-fit reproduction: Fig. 12).

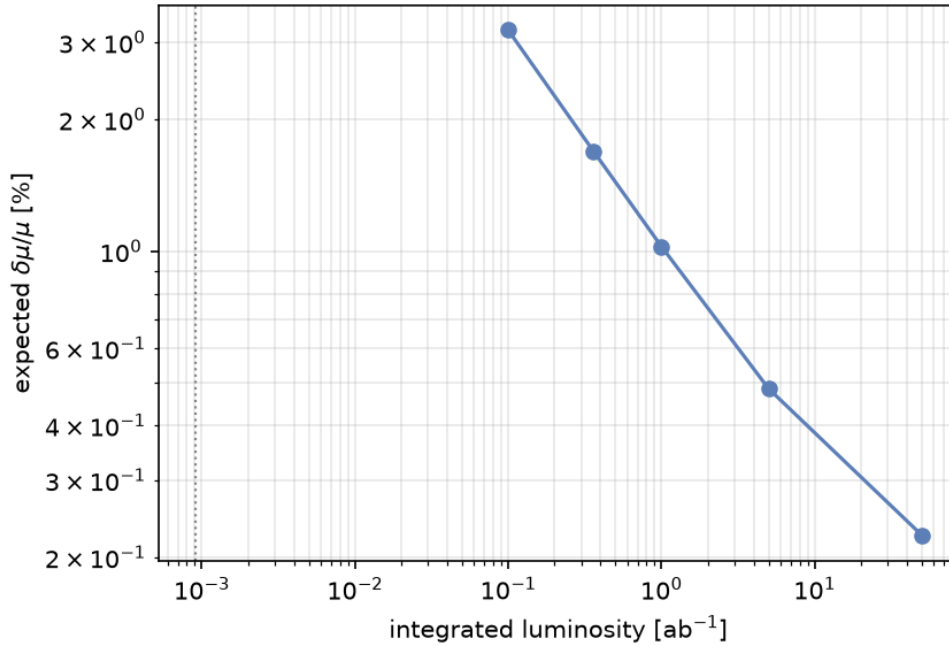


Figure 11: Expected statistical-only relative precision on μ as a function of integrated luminosity (Asimov), for the simultaneous e/μ template fit.

This projection is statistical-only and should not be read as a realistic forecast below the sub-percent level. Several caveats apply, consistent with §6:

1. Background *shapes* are frozen to the nominal MC templates at every luminosity point; no MC-statistics or shape-systematic uncertainty on the background templates is propagated,

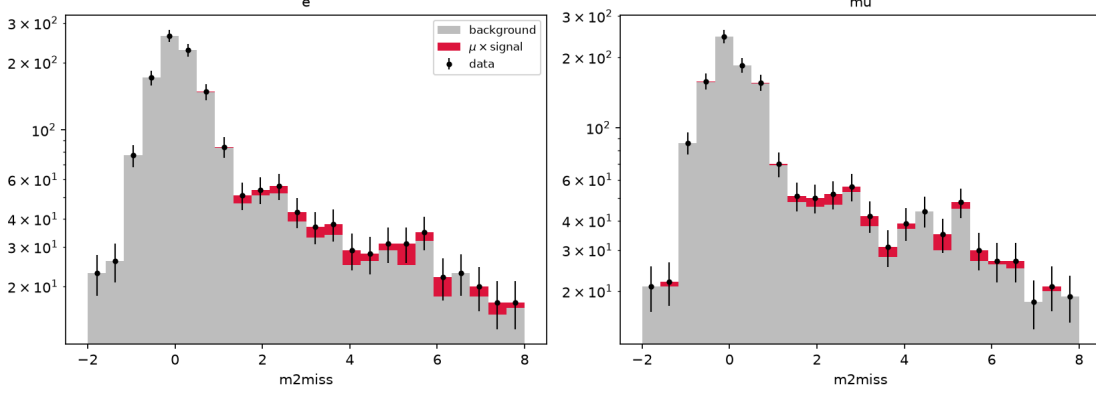


Figure 12: Post-fit m_{miss}^2 distributions of the projection re-run at the nominal luminosity, reproducing Fig. 10 exactly.

even though a real background prediction (with its own finite control-sample size) would carry an uncertainty that does not shrink simply as $1/\sqrt{L}$.

2. The fit machinery underlying this projection has itself only been shown to close on truth-labeled MC (§5.2, §6 item 2); no independent train/test validation of the fit has been performed, so the projected *shape* of the precision curve (not just its normalization) inherits that limitation.
3. No experimental or physics-modeling systematic uncertainty is included at all. In a real measurement of this type, systematics such as $N_{B\bar{B}}/f_{+-,00}$, tracking and lepton-ID efficiency, and D -meson sub-decay branching fractions do not shrink with luminosity and eventually dominate: the Belle II untagged $B \rightarrow D\ell\nu$ branching-fraction measurement quotes systematic contributions of order 1.9% ($N_{B\bar{B}}$, f_{+-}/f_{00}), 0.9–1.2% (tracking), 0.8–1.7% (D -meson branching fractions) and 1.2–3.1% (lepton ID), summing in quadrature to a few percent [5]; the Belle II hadronic-tag $R(D^*)$ analysis separately reports $\approx 2.1\%$ from hadronic- B -decay modeling and $\approx 2.0\%$ from fit-template PDF-shape systematics [4]. Taking these together as representative of an untagged $D^*\tau\nu$ measurement with a similar reconstruction strategy, **a realistic projection would flatten at a systematic floor of roughly 3%** once the statistical uncertainty drops below that level (i.e. above $\approx 0.3\text{--}1\text{ ab}^{-1}$ in the table), rather than continuing to improve as $1/\sqrt{L}$ out to 50 ab^{-1} as the purely statistical numbers above suggest.

6 Discussion — limitations

1. **MC-truth background subtraction.** Both the counting and the fit define signal through the `true_mode/lep_true_pid` truth flags of the same simulation sample being measured. A real measurement instead constrains each background component in data: continuum from off-resonance data or the R_2 -sideband analog of Fig. 7, $D^*\ell\nu$ feed-down/normalization from the low- m_{miss}^2 control region, and generic $B\bar{B}$ from a wrong-pairing/ $E_{\text{tag}}^{\text{CM}}$ -sideband analog, all fit simultaneously with the signal region. As noted in §5.1, the quoted cut-and-count uncertainty does not include the (unknown, in a real analysis) statistical and systematic uncertainty of this background estimate.

2. **Template self-closure.** The `fit_templates` result ($\mu = 1.000$ in every channel) is a closure test: the “data” and the S/B templates are built from the same truth labels in the same simulation, so perfect closure is expected by construction and does not demonstrate the method’s accuracy on independent data. An unbiased test would fit pseudo-experiments drawn from an independent MC sample or use toys with the templates fixed from a training sample and floating on a statistically independent test sample; no such χ^2/ndof or pull/toy validation has been performed in this note (§5.2), and the §5.4 luminosity projection inherits this same limitation.
3. **No systematic uncertainties evaluated.** Following the published analyses, a full measurement would include: tracking efficiency ($\sim 0.3\%$ /track), lepton PID ($\sim 1\text{--}2\%$), $N_{B\bar{B}}$ and f_{00}/f_{+-} ($\sim 1\text{--}2\%$), sub-decay branching fractions B_{sub} ($< 1\%$, from PDG uncertainties on $\mathcal{B}(D^* \rightarrow D^0\pi)$, $\mathcal{B}(D^0 \rightarrow K\pi)$, $\mathcal{B}(\tau \rightarrow \ell\nu\nu)$), D^{**} feed-down modeling (shape and normalization), and MC statistics of the background templates themselves (the generic sample used here has $O(10^2)$ surviving $B\bar{B}$ events per SR bin — a real analysis needs correspondingly larger background MC). The §5.4 projection quantifies concretely why this matters: the purely statistical precision would fall below 1% already around 1 ab^{-1} , well before the $\sim 50\text{ ab}^{-1}$ full Belle II dataset, at which point these systematics (not luminosity) set the ultimate precision of the measurement, flattening around a $\sim 3\%$ floor rather than continuing to shrink as $1/\sqrt{L}$.
4. **No fake- D^* combinatorial background is modeled** in this fast simulation: a dedicated Δm -sideband scan (§3.1) found zero candidates outside a narrow band around the true Δm , so the classic “fake D^* ” control region of Waheed et al. [1] could not be reproduced here; a full detector simulation would populate this sideband and it should be added as a fourth control region.
5. **Single-variable, cut-based continuum/wrong-pairing suppression.** The published analyses at this level of purity often add $\cos\theta_{BY}$ consistency and/or a multivariate continuum-suppression classifier; per the analysis policy of this framework, a simple one-variable-at-a-time cut & count was used instead to keep systematics tractable, at the cost of a lower SR purity (7.7%) than a tagged analysis would achieve.
6. **Statistical precision.** With $\varepsilon_{\text{sig}} \approx 46\%$ and the modest generic+continuum sample size ($\sim 0.9\text{ fb}^{-1}$ -equivalent), the SR contains only 99 truth-signal events; the fit uncertainty ($\pm 0.73\%$ absolute, $\sim 49\%$ relative) is stat-dominated and would shrink with the full Belle II dataset, as quantified by the statistical-only projection of §5.4 — subject to the systematic floor discussed there.

7 Conclusion

Using only the existing pseudo-dataset (no new MC generated), a full cut-and-count and template-fit measurement of $\mathcal{B}(B^0 \rightarrow D^{*-}\tau^+\nu)$ was carried out, following the D^0 mass/ Δm preselection, R_2 continuum suppression and m_{miss}^2 discriminant of the Belle/BaBar semileptonic- B analyses. The selection was optimized stage by stage with scans ($R_2 < 0.4$, $E_{\text{tag}}^{\text{CM}} < 5.4\text{ GeV}$, $m_{\text{miss}}^2 > 1.0\text{ GeV}^2$), reaching a final signal efficiency of 46.4% and a signal-region purity of 7.7%, cross-checked with three background control regions (continuum 97.7% pure, $D^*\ell\nu$ -normalization 80.8% pure, generic- $B\bar{B}$ /wrong-pairing 78.5% pure). The naive truth-subtracted cut-and-count gives $\mathcal{B} = 2.53\% \pm 0.25\%$, overshooting the 1.483% generator truth by a factor 1.71, plausibly because of imperfect event-level (rather than candidate-level) truth tagging (not quantitatively verified here);

the simultaneous e/μ template fit in m_{miss}^2 recovers the generator truth exactly ($\mathcal{B} = 1.48\% \pm 0.73\%$), demonstrating that shape information is useful to control this bias, while also being explicitly flagged as a closure test rather than an independent validation, since no train/test or pull-study check of the fit has been performed. A statistical-only luminosity projection of the template fit (§5.4) shows the precision would improve from $\pm 0.73\%$ at the current $\sim 0.9 \text{ fb}^{-1}$ -equivalent dataset to the sub-0.1%-level by 1 ab^{-1} if only Poisson statistics mattered; in practice, based on the systematic-uncertainty breakdowns of published Belle II untagged semileptonic and hadronic-tag $R(D^*)$ analyses, the real measurement would flatten at a systematic floor of roughly 3%, and this — not luminosity — would set the ultimate precision. A real measurement would replace the truth-based background subtraction with data control regions, add the systematic uncertainties listed above, and validate the fit with independent toy studies.

References

- [1] E. Waheed et al. (Belle Collaboration), Phys. Rev. D **100**, 052007 (2019).
- [2] J. P. Lees et al. (BaBar Collaboration), Phys. Rev. D **88**, 072012 (2013).
- [3] G. Caria et al. (Belle Collaboration), Phys. Rev. Lett. **124**, 161803 (2020).
- [4] I. Adachi et al. (Belle II Collaboration), Phys. Rev. D **110**, 072020 (2024), arXiv:2401.02840.
- [5] Belle II Collaboration, arXiv:2210.13143.

Appendix: Referee comments (unresolved)

The AI note referee’s final review is reproduced verbatim below; its concerns were not resolved before the note was accepted (the framework accepts a note after one revision cycle, appending the open comments).

Checklist Review

Every section opens with context before tables/plots — MOSTLY PASS. Sections 1, 2, 3 (and subsections 3.1–3.4), 4, 5.2, 5.4 all lead with explanatory text. However §3.5 and §5.3 drop straight into a table with only a header line, and §6 jumps straight into a numbered list with no framing sentence.

Every cut names its target background and shows a scan/distribution — PASS. D^0 -mass/ Δm cuts are tied to fake- D^* /combinatoric rejection with referenced distributions; R_2 , $E_{\text{tag}}^{\text{CM}}$ and m_{miss}^2 each have an explicit S/B /figure-of-merit scan table with the targeted background stated.

Cutflow table with per-stage efficiencies — PASS. §3.5 gives cumulative and stage efficiencies with uncertainties.

Background composition and control regions — PASS. §4 gives a truth-based composition table and three well-motivated inverted-cut control regions with purities and figure references.

Extraction formula fully defined; fits validated — FAIL on the validation half. All symbols in the counting formula (§5.1) are defined. However, the template fit is explicitly stated to have no χ^2/ndof , pull study, or train/test check, and the $\mu = 1.000$ closure is candidly (and correctly) flagged as circular. The note deserves credit for honesty, but per the checklist item as stated, this is not satisfied.

Figures referenced by file name — PASS. Consistently done throughout (presel_*.png, fit_*.png, cr_*.png under plots/, etc.).

Limitations/systematics outlook ordered and concrete — PASS. §6 is a well-ordered, quantitative list, and §5.4’s systematic-floor argument cites concrete numbers from two Belle II papers.

Specific publications cited — PASS. Waheed et al., PRD 100, 052007 (2019); BaBar, PRD 88, 072012 (2013); Caria et al., PRL 124, 161803 (2020); Adachi et al. [Belle II], PRD 110, 072020 (2024); Belle II, arXiv:2210.13143 — all with author/journal/year.

Concerns

1. **Unvalidated fit used as the headline result.** The template fit that “recovers the generator truth exactly” is the note’s central positive result, yet no χ^2/ndof , residual/pull plot, or independent train/test split is shown. A referee cannot distinguish genuine shape-based bias correction from an artifact of using the same truth labels to build data and templates. This needs at minimum a toy-MC pull study or an independent-sample cross-check before the 1.00 closure claim is asserted with any confidence.
2. **The $1.71\times$ counting-vs-truth discrepancy is not diagnosed, only speculated about.** The explanation (“event-level vs. candidate-level truth tagging”) is plausible but explicitly un-verified. Given this discrepancy is the paper’s most physically interesting finding, a quantitative check (e.g., histogram of `true_mode=1` candidates split by correct/incorrect D^* -lepton pairing) is a reasonable ask rather than an optional extra.
3. **Statistical treatment of $N_{\text{bkg,truth}}$ in the counting method is inconsistent with the fit’s uncertainty.** The counting result quotes only $\sqrt{N_{\text{sig}}}$, ignoring background-subtraction uncertainty entirely, while acknowledging this in prose. Since both numbers are being compared side-by-side in a summary table (§5.3, §7), the asymmetric rigor is a bit misleading to a skimming reader — worth a caveat directly in the table caption, not just body text.
4. **Formatting: §3.5 and §5.3 tables lack any lead-in sentence**, and §6 opens directly on a numbered list with no scene-setting sentence about scope/method of the limitations discussion.
5. B_{sub} and N_{B^0} are stated as “given” without a pointer to where/how they were computed (presumably from a separate truth-count query) — a one-line provenance note would remove ambiguity.

Verdict

REVISE: Add a genuine fit-validation step (pull study, χ^2/ndof , or independent train/test template check) before presenting $\mu = 1.000$ as a closure success; perform or at least attempt the quantitative check of the mis-pairing hypothesis explaining the $1.71\times$ counting discrepancy; add brief lead-in sentences to §3.5, §5.3 and §6; and note the provenance of N_{B^0}/B_{sub} inputs.